

How To Build an Efficient Personal Knowledge Management System



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I believe that making an effort to establish a personal **knowledge management system** can be beneficial to everyone. Nowadays, we all face with too much **fragmented information**. We have to learn how to control the information by transforming quality content into knowledge and discarding momentary and useless information.

Every morning, when I wake up, open my mobile phone or laptop, enormous information rushes towards me. Originally, I feel so excited for I can instantly access the information on any topic from anywhere. It seems that I can obtain any knowledge in any field and solve any problem in the world.

I have read a lot of articles, posts and so on, and also saved a lot of them to my hard disk. Unfortunately, I gradually realized that although I spent plenty of time on various information, I did not become more knowledgeable. And sometimes, I even cannot remember that I have studied a topic despite the fact that I have stored many articles regarding the topic on my computer.

It is a sad truth that the information I have obtained just stored on the computer instead of saving in my brain and becoming my **knowledge**,

which means that I still cannot freely recall it, organize it and use it whenever and wherever. Therefore I have to build a *system* which can help me to translate information into knowledge, and I did.

What is Knowledge Management

Knowledge Management is one of the hottest topics today in both the industry world and information research world. In our daily life, we deal with huge amount of data and information. Data and information are not knowledge until we know how to dig the value out of it. This is the reason we need knowledge management. Unfortunately, there's no universal definition of knowledge management, just as there's no agreement as to what constitutes knowledge in the first place.

There are over a hundred published definitions of knowledge management (Dalkir 2011) and perhaps the widely accepted one is developed by the Gartner Group and Bryant Duhon:

“Knowledge management is a discipline that promotes an integrated approach to identifying, capturing, evaluating, retrieving, and sharing all of an enterprise's information assets. These assets may include databases, documents, policies, procedures, and previously un-captured expertise and experience in individual workers.” (Duhon, 1998)

Why We Need to Build a Personal Knowledge System

Although KM generally refers to achieving **organizational** objectives or **business** result, I mainly focus on **personal** KM in this article to help us develop ourselves more and gain more knowledge.

After building my KM system, I keep improving and modifying it, which makes me become more efficient and help me completely solve the following issues:

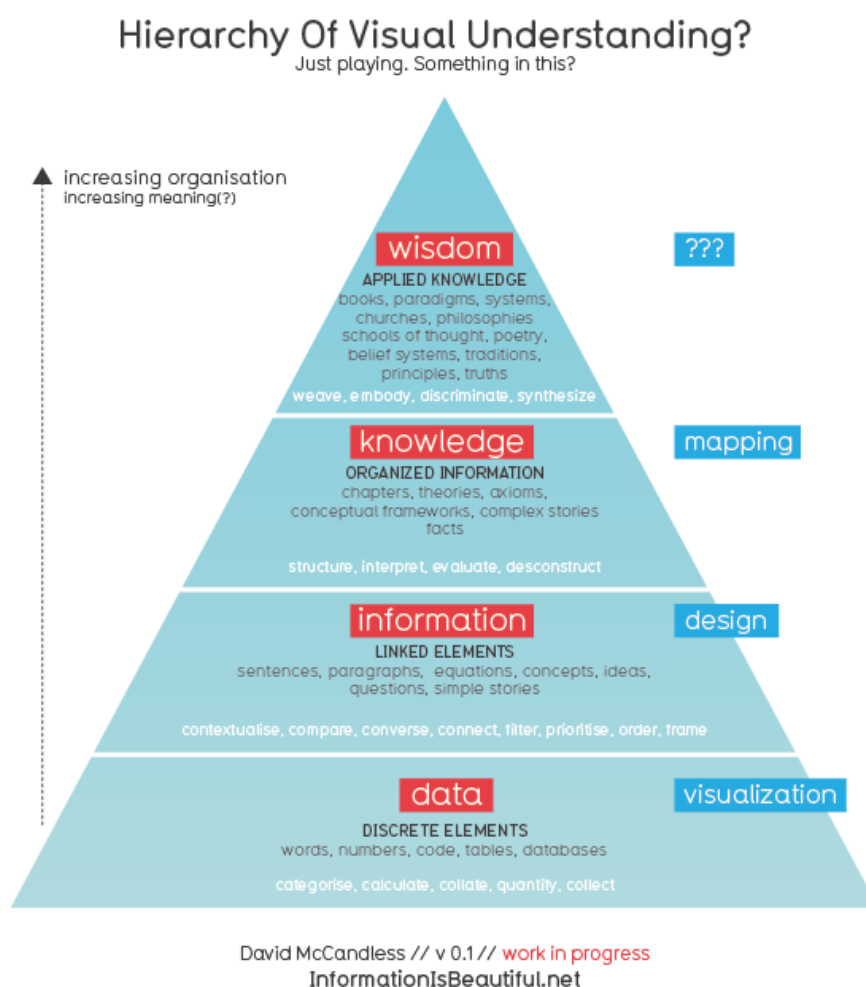
1. Amass information without creation.

As I have mentioned before, every day I read a lot of articles or posts from websites, social media, and Google search. When I found an excellent article, I will promptly save it to Evernote, a digital notebook I have used before, for further reference. After many years accumulation, there are thousands of notes in my Evernote notebook. However, I did not read most of them for second times after saving them.

1. No output, no sharing.

For example, when I was dealing with a problem, I always referred to multiple resources and even did certain experiments to conclude the final solution. After the task had been done, I did not organize all the information regarding the solution and write down how I found it. Also, I did not share it with someone else. After a time, I will forget all about these: the references, the experiments, and the solution. All my efforts do not achieve their maximum value.

The Key To A Good KMS



The above picture is a DIKW (abbreviation of **D**ata, **I**nformation, **K**nowledge, **W**isdom) pyramid from David McCandless, which illustrates the hierarchy of data, information, knowledge and wisdom. It is important to note that there is a definite **boundary** in the middle of the pyramid: Data and information are what we **collected**, Knowledge and wisdom are what we **created**.

The essential part of a good KMS is **output**. Reading books, watching videos and attending lectures are all activities for acquiring information. Even though books and lectures belong to wisdom or knowledge in the DIKW model, they are the authors' wisdom and lecturers' knowledge. They are not ours. The best way to transform the information into our knowledge is to **interpret** something new to us based on what we have learned and **share** them.

Interpreting helps us organize and digest information. Therefore it can translate information into our embedded knowledge. And sharing is not only a contribution to the information age but also our motivation to keep outputting better works, which is the reason why I believe that information becomes knowledge via organizing and sharing.

Working Flow

Basic Flow



Above figure illustrates the primary working flow of my KMS, which is “**Collect, Organize, and Output.**” The three phases correspond to the DIKW model. After organization, the information that we collected transforms to our knowledge. Then if we applied the knowledge in practice, such as writing an article or resolving a problem, it would eventually become our wisdom.

The figure also shows the iOS apps I currently use in each phase. Although any app would be changed in the future if I found a better one, they are currently the best choice for me after testing many similar apps.

Storage

OneNote is the most important application of this system. It is the central storage and the perfect note taking solution. All information which is processed by me is classified and saved in OneNote. **EverNote** acts as a container for collected information and the final storage for raw materials. Materials which I do not study them and write notes on them are called “raw material,” I only keep them for further reference. EverNote used to be my primary note app for a long time, while it is also a fantastic software, I finally moved to OneNote because of following reasons:

- Windows version OneNote has better performance than EverNote. If an EverNote notebook contains too many notes, say more than one thousand, the performance would become terrible. I cannot scroll the notebook smoothly even it is running on my i7 CPU with 16G RAM laptop.
- Every OneNote's page is an endless paper, which is perfect to complement additional content to a note.

Collect

We face with two types of information: Active information and passive information. Active information is what we intend to acquire, such as googling a keyword. By contrast, passive information is pushed to us by its' senders, such as twitters, posts and so on.

In the collection phase, the primary task is to **judge which information should be kept for further reading**. And it is also important that do not spend too much time on browsing the passive information.

Every day, I spend about thirty minutes browsing rapidly the articles which I have subscribed in LinkedIn, Facebook, Twitter and Feedly. If I found an interested one, I would send it to *Pocket* (If this post is in LinkedIn, Facebook, Twitter) or save it (in *Feedly*) for later reading.

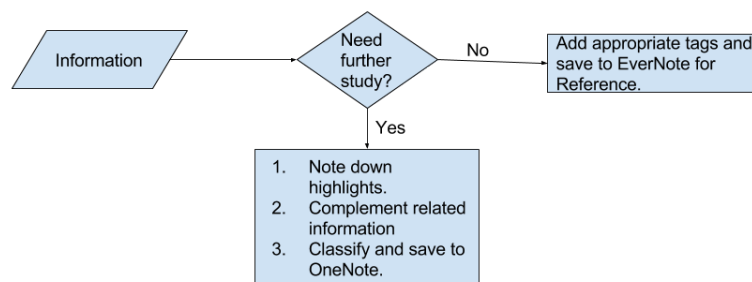
Workflow is a powerful automation tool which can “connect apps and actions together to automate things you do on your device.” I use it to extract useful information from Safari by:

- Converting the whole web page to PDF document and save it to *Dropbox* for further processing.
- Excerpting snippets from a web page and appending them to *EverNote* notes.

I digitize paper documents via *Scanner Pro* and save them to *Dropbox*, and use *Duokan* to read ePUB books because it can automatically save the book's highlights or notes to *EverNote*.

After this phase is finished, all the information which need to be processed in next phase is stored in **Pocket, Feedly, Dropbox and EverNote**.

Organization



For each of the collected material, we need to decide whether we should study it or just keep it for further reference. The decision depends on subjects on which we are currently focusing. For example, if I am currently concentrating on web development, an article named “10 tips to build a fast HTML 5 app” is definitely worth to read carefully. And “Healthy dinner recipes”, I am not interested in cooking, but these recipes seem perfect and maybe someday I will need to prepare a feast, so it would be better to save it in my “Reference” notebook.

Organization mainly includes classification and complement.

Besides highlighting and noting down the important points in an article, it is also necessary to expand on certain contents or words to help us understand more deeply. If a post contains a term “DES”, for instance, which I do not know its meaning, I will google it and note down the explanation.

After processing a material, I should save it to an appropriate location and tag it, in other words, classify it into the proper category. In OneNote, I am using three levels for classification: Notebook, Section and Tag. Regarding how to build a personal note system, this is another topic which is worth writing the other article to describe it in detail. Simply put, the principle of my note taking solution is that the notes should be obtained or consulted again easily.

The most important step is to **review** the organized materials periodically. Without review, our hard disk would become a landfill instead of a treasure trove of knowledge. I do review weekly. During the review, I always ask myself following questions about an organized note:

- Is it relevant to my current focus?

If yes, then I need to put more effort on it.

- It can be used to solve which problem?

This question helps create a relationship between the knowledge and problems. We can keep in mind that which problems can refer to these notes.

- Does it belong to which topic or project?

Classify the note to existing topic or create a new topic if necessary. For example, before I began to write this post, I created a topic "Knowledge Management System ." So related articles or pictures will be classified to this topic.

There are not too many apps needed in this phase. Following are the three top apps I frequently use:

- MarginNote Pro

This is my ideal app for PDF documents studying, annotating and noting. It combines Mindmap and Outline for note organization and provides Flashcard for memorizing notes. It has a free version named "MarginNote" which you could try it.

- MindJet Maps

A free mind map app with standard functions. I only read mind maps on my iPad and rarely edit them, so MindJet fulfills my needs. If you need to edit mind map frequently, I would not recommend it.

- OneNote

A powerful and cross-platform note taking solution. Each page likes an endless paper, which makes it extremely convenient for complementing new contents to an existing note.

After the organization phase, all the organized and processed information, and information which is being organized and processed, are saved in **MarginNote Pro** and **OneNote**.

Output

As I have mentioned before, the **output** is an essential stage of knowledge management. Output, such as writing an article, telling a story or delivering a speech, compels us to structure and reorganize the knowledge that we have learned, and to depict it in an explicit way which others could easily understand. After this stage, we grasp the knowledge because we have known it, understood it and used it.

Apart from the Microsoft Office and Sway, I use **Ulysses** for most of my writing tasks. Ulysses is a decent *Markdown* editor app which can make me concentrate on typing and enjoy writing.

Conclusion

Collection, organization and output compose a whole system of knowledge management. A lack of any one of these three parts will cause us getting half the result with double efforts in learning.

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Axton Liu writes at axtonliu.com, where he shares technical based ideas and what he has learned so far. You can also follow [his blog](#) to never miss an update.

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